EPSR-test-ex10 Two X-ray data sets (nxfile = 2)

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Extends ex9 to two X-ray data sets, the `MXFILE = 2` limit of the `epsr` engine [1,2], with the alternative one-number header of the X-ray weights file and non-zero modified atomic form factors. X-ray data are weighted by Q-dependent atomic form factors instead of constant scattering lengths, so the weights, the self-scattering background and the EP update all take separate code paths.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `get_wts` (two X-ray files, one-number header decoded into density and normalisation flags), `wxtype` (modified form factors), `wxcreate`
- `epsr_standalone_rev3.f`: `runepsr17` and `ptgnsq` X-ray branches for `nxfile` = 2, `interp`
- `epsr_standalone_rev3.f`: `inita` (`iinit 1`)

Rust (port, rust/epsr-core/src/):

- `weights.rs`: `State::get_wts`, `read_xray_weights`, `wxtype`, `wxcreate`
- `epsr.rs`: `Epsr::iterate`; `potential.rs`: `State::ptgnsq`
- `structure.rs`: `interp`, `State::inita`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [3] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). Data set 1 is the neutron data of deuterated DMSO in D2O (SANDALS run SLS03863) [4]. Data sets 2 and 3 are treated as X-ray data with Waasmaier-Kirfel form factors [5]; their data files are byte-identical copies of the null-water (SLS03871) and H2O (SLS03850) neutron data, so the case tests code paths, not a physical fit. Weights file A has header `-2`, B has `3` (ex9: `1 2 1`), and modified form factors $f(Q)\cdot(1 - (\mathrm{alpha}/f(0))\cdot\exp(-Q^2/(2\cdot\mathrm{delta}^2)))$ are switched on (alpha 0.5, delta 2.0 for O/O1 in A; 0.3, 1.5 for H/H1 in B).

Inputs

As ex9, but data sets 2 and 3 in the `.EPSR.inp` use `D6DMSO_HDO_xray.mint01` + `DMSOWaterA.XWTS.wts` and `DMSOWater_xray.mint01` + `DMSOWaterB.XWTS.wts` (variants of the ex9 weights file with changed header and alpha/delta values); `iinit 1`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Two chained passes (pass 1 with `iinit 1`, pass 2 continuing). `get_wts` decodes header `-2` to normalisation by $\langle f \rangle^2$ of the independent-atom form factors (`datanorm` 2) and `3` to normalisation by $\langle f^2 \rangle$ of the modified form factors, so both normalisation branches of `wxcreate` and the MAFF branch of `wxtype` run. `wxcreate` rewrites `X-raywts.dat` and `X-rayf0x.dat` for each X-ray set, so the files left are those of set 3; they and the 27 standard outputs (29 files) are identical after both passes.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] D. Waasmaier and A. Kirfel, "New analytical scattering-factor functions for free atoms and ions", Acta Cryst. A51, 416-431 (1995).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.